THE LOG-SINE CONSTANT C_g AS THE SHARP CONSTANT IN THE RAMSEY CONTRAST OF A THERMAL CAVITY MODE

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Abstract. A two-level probe coupled dispersively to a cavity mode in a thermal state with mean occupation $\bar{n}$ has the well-known Ramsey contrast $V(u) = (1 + 4\bar{n}(\bar{n}+1)\sin^2 \pi u)^{-1/2}$, where u is the delay in units of the period of the dispersive phase. At the half period it equals $1/(2\bar{n}+1)$, which already gives $\bar{n}$ in this ideal model, and no new measurement is proposed. We study the geometric mean of the contrast after the first return, $\mathcal{V}_s = \exp\left(s \int_1^\infty \log V(u)\, u^{-s-1} du\right)$, $s > 0$. For every positive occupation, $p_0 < \mathcal{V}_s < e^{-sG(s)} p_0$, where $p_0 = 1/(\bar{n}+1)$ is the vacuum probability and $G(s) = \int_1^\infty \log|2 \sin \pi u|\, u^{-s-1} du$. Both constants are the best possible, and for $s = 1$ the upper one is $\exp(C_g)/2 = 1.05195\ldots$, with $C_g = -\int_1^\infty \log|\sin \pi u|\, u^{-2} du$. For every photon-number distribution of compound Poisson type, which includes thermal, Poisson and displaced thermal states, $p_0 < \mathcal{V}_1 \le p_0^{1-c}$ with $c = 0.0409\ldots$, whereas for a given p_0 the half-period contrast can take every value in $[p_0^2, 1]$. We also bound the effect of damping and of a finite range of delays, prove a sum rule, and relate G and the transform A of the delays before the first return to the Riemann zeta function. The functions G and A are shown with their computed zeros.

Contents

1. Introduction 1
2. The model and the contrast 3
3. The coefficients of the contrast and the transform after the first return 5
4. The sharp bound 7
5. Damping, intrinsic dephasing and a nonlinear shift 12
6. The sum rule and the harmonic delays 14
7. The transforms after and before the first return and their zeros 15
8. Conclusions 21
References 21

1. Introduction

In a superconducting quantum processor each qubit is read out through a microwave resonator, and a few thermal photons left in that resonator are enough to dephase the qubit [12, 15]. The interaction responsible is dispersive: the transition frequency of the qubit shifts by a fixed amount for every photon in the mode [1]. The same interaction turns the qubit into a probe of the photon statistics. This note is about one exact property of that probe when the mode is thermal. It does not introduce a new method of measurement, and we first state what is known.

Two families of methods are established. The first works in the frequency domain. When the shift per photon exceeds all linewidths, the transition of the qubit splits into one line for each photon number. This was predicted in [8] and observed in [13], where the strengths of the lines distinguished a coherent field from a thermal one. A pulse that is selective on one line maps a

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chosen photon number onto the state of the qubit [9], and Rabi oscillations driven on the separate lines are used to determine the occupation of the mode [15].

The second family works in the time domain and is older. Brune et al. proposed a quantum non-demolition measurement of small photon numbers by the phase-sensitive detection of atoms coupled dispersively to the field [3]. A Ramsey sequence on the dispersively coupled probe is the usual form of this measurement: each photon number gives a different phase of the probe [9]. With a delay of half the period of the dispersive phase, the Ramsey sequence maps the photon-number parity of a cavity onto the qubit [16]. The coherence of the probe after a delay is the characteristic function of the photon number. For a thermal mode in contact with a bath it is known in closed form: the result is due to Dykman and Krivoglaz [6] and was derived again, including a drive, by Clerk and Utami [4]. Its long-time limit gives the dephasing rate that is used to interpret coherence times [12], and Ramsey experiments with injected thermal photons test it quantitatively [15]. Thermal and coherent photon noise have been distinguished through the noise spectrum seen by the qubit [18], and a superconducting circuit has been used as a primary thermometer of the thermal occupation of propagating microwave modes [14]. More generally, the coherence of a probe is a partition function of the system to which it is coupled, continued to a complex parameter [17], and such functions and their zeros have been measured [11].

In a mode without damping the photon number is constant during the sequence, and the contrast of a thermal state is the elementary closed form (2.4), the case without damping of [6, 4]. It is periodic in the delay and returns to 1 at every multiple of the period π/χ of the dispersive phase, where 2χ is the shift of the transition frequency per photon; we call these instants the returns. At the half period the contrast is the modulus of the mean parity, $1/(2\bar{n}+1)$, where $\bar{n}$ is the mean occupation. In this ideal model a single Ramsey delay therefore determines $\bar{n}$ exactly.

Nothing in the preceding paragraphs is new here: neither the Ramsey method, nor the contrast of a thermal mode, nor the fact that this contrast determines the occupation. Within the thermal model the half-period contrast is the simplest way to obtain $\bar{n}$, and the functional studied in this note does not improve on it. The subject of the note is a global functional of the known contrast, and its results are exact statements about that functional: best constants, a comparison with the half-period contrast when the photon statistics is not known, a bound for the effect of damping, a sum rule and the structure of a Mellin transform.

The objective is to determine how the contrast accumulated after the first return depends on the occupation of the mode, when it is averaged geometrically over the delay with a power weight, to find the best constants in that dependence, and to state precisely what this functional measures that the half-period contrast does not.

The answer to the first part is Theorem 4.1. For every exponent $s > 0$ of the weight and every positive occupation, the geometric contrast $\mathcal{V}_s$ lies strictly between the probability p_0 that the mode is empty and $e^{-sG(s)}p_0$, and neither constant can be improved. Here G is the transform (3.12) of $\log|2\sin\pi u|$ on $[1,\infty)$. For the weight u^{-2} the upper constant is $\exp(C_g)/2$, where C_g is the log-sine constant (3.14). Corollary 4.2 turns the bound into a bracket for the occupation, and Corollary 4.3 combines it with the half-period contrast into a relation without parameters that holds for every thermal state and fails for coherent states of mean photon number at least 0.3.

The answer to the second part is Proposition 4.4. For every photon-number distribution of compound Poisson type, which includes the thermal, the Poisson and the displaced thermal distributions, the geometric contrast after the first return lies between p_0 and $p_0^{1-c_1(1)}$ with $c_1(1) = 0.0409\ldots$, that is, the ratio $\log\mathcal{V}_1/\log p_0$ lies in $[1-c_1(1),1) = [0.959\ldots,1)$ without any further hypothesis on the statistics. This is an exact property of the ideal contrast, not an experimental error bar. The half-period contrast is the modulus of the mean photon-number parity; for a given p_0 it can take any value between p_0^2 and 1 in the same class, and it gives p_0 or $\bar{n}$ only through the thermal hypothesis. A mode that holds residual coherent photons together with thermal ones, the situation analysed in [4, 18], is of this kind. This is the only comparative statement of the note.

Theorem 4.5 gives the expansion of $\mathcal{V}_1$ at high occupation with an explicit remainder, and Proposition 4.6 bounds the effect of a finite range of delays. Section 5 treats the idealisations. Proposition 5.1 bounds the effect of damping of the mode in the rate equations of [15, 4]: the coherence changes by at most $4\bar{n}(\bar{n}+1)\kappa t$. The condition it puts on the geometric contrast grows as $\bar{n}^3$, and it shows that the half-period contrast is less affected by damping. Proposition 6.1 shows that the integral of $\log V$ over all delays with the weight u^{-2} is exactly $-\pi^2\bar{n}$. Proposition 6.2 shows that the product $\mathcal{P}(N)$ of the contrasts at the harmonic delays N/k satisfies $\mathcal{P}(N)^{1/N} \to \mathcal{V}_1$. Section 7 treats the function G at complex exponents together with the companion transform A of (7.2), which collects the delays before the first return, proves the identity (7.4) that links them to the Riemann zeta function, and shows both functions with their computed zeros.

On the mathematical side, the classical log-sine integrals are integrals of powers of $\log|2\sin(\theta/2)|$ against powers of θ over a finite interval, and their evaluation in terms of zeta values and polylogarithms has a long history; see [2] and the references there. The function G is of a different kind: it is a Mellin transform of $\log|2\sin\pi u|$ over the half-line $[1,\infty)$, and C_g is $\log 2$ minus its value at $s=1$. Mellin transforms over a half-line that represent the Riemann zeta function are classical, and the identity (7.4) is of this type. We do not claim it as new; we prove it and use the way in which the first return splits it into G and A. The sum rule (6.1) is an elementary consequence of the Dirichlet integral. To our knowledge the bounds (4.3) and (4.9), and the role of C_g as the sharp constant of the first, have not been stated before.

All statements except those of Section 5 concern the ideal model (2.1): a single mode without damping, a probe without intrinsic dephasing, and a dispersive shift that is exactly linear in the photon number. The returns of the contrast to 1 rest on these three idealisations. The geometric contrast is an integral over unbounded delays, and it averages the logarithm of the contrast, so it gives weight to the delays where the contrast is small, down to $1/(2\bar{n}+1)$, where a measured contrast is most affected by noise and by damping. No experimental advantage over the existing methods of thermometry is claimed beyond Proposition 4.4.

The proofs are elementary and self-contained. Four classical facts are quoted: from [5] the product for the sine, the partial fractions of the cotangent and the Dirichlet integral, and from [7] the representation of the thermal state as a Gaussian mixture of coherent states. The figures and tables are computed in double precision or in multiple precision arithmetic and illustrate the statements; the scripts that produce them accompany the note. Table 1 lists the notation.

2. The model and the contrast

Let $\hat{a}$ be the annihilation operator of the mode and $\hat{\sigma}_z$ the Pauli operator of the probe, and let

$$\hat{H} = \omega_c\,\hat{a}^\dagger\hat{a} + \frac{\omega_p}{2}\,\hat{\sigma}_z + \chi\,\hat{a}^\dagger\hat{a}\,\hat{\sigma}_z, \tag{2.1}$$

in units in which the reduced Planck constant is 1, with real ω_c, ω_p and $\chi > 0$. The mode is in the thermal state of mean occupation $\bar{n} > 0$, in which the photon number $n = 0, 1, 2, \ldots$ has probability

$$p_n = (1-q)\,q^n, \qquad q = \frac{\bar{n}}{\bar{n}+1}, \qquad p_0 = 1-q = \frac{1}{\bar{n}+1}. \tag{2.2}$$

The probe is prepared in an equal superposition of the two eigenstates of $\hat{\sigma}_z$. The Hamiltonian is diagonal in the photon number, and the two eigenstates of $\hat{\sigma}_z$ with n photons have energies that differ by $\omega_p + 2\chi n$, so that the transition frequency of the probe shifts by 2χ per photon. After a time t the off-diagonal element of the state of the probe is therefore its initial value multiplied by $e^{-i\omega_p t}\sum_{n\ge0} p_n e^{-2i\chi tn}$. We measure the delay in units of the period π/χ of the second factor, $u = \chi t/\pi$, and put

$$D(u) = \sum_{n\ge0} p_n\, e^{-2\pi i n u}, \qquad V(u) = |D(u)|, \qquad u \in \mathbb{R}. \tag{2.3}$$

TABLE 1. Notation.

symbol	meaning
$\hat{a}$, $\hat{\sigma}_z$, $\hat{H}$	annihilation operator of the mode, Pauli operator of the probe, Hamiltonian (2.1)
ω_c, ω_p, χ	frequencies of the mode and of the probe, dispersive coupling; the transition frequency of the probe shifts by 2χ per photon
t, $u = \chi t/\pi$	delay, and delay in units of the period π/χ
n, p_n	photon number and its probability (2.2)
$\bar{n}$, $q = \bar{n}/(\bar{n}+1)$, $p_0 = 1-q$	mean occupation, ratio of the thermal distribution, vacuum probability
$\Lambda = 4\bar{n}(\bar{n}+1)$, $L = \log(2\bar{n}+1)$, $\tau = \log(1/q)$	auxiliary thermal parameters
$D(u)$, $V(u) = \lvert D(u)\rvert$	coherence and contrast of the probe, (2.3)
$\Phi_q(u)$, $h(u)$	$\log\lvert 1-qe^{2\pi i u}\rvert$ and $\log\lvert 2\sin \pi u\rvert$, (2.5) and (3.12)
$s = \sigma + i\theta$	exponent of the weight u^{-s-1}
$c_m(s)$, $m \geq 1$	coefficients (3.1)
$G(s)$, $G_q(s)$, $A(s)$	transforms (3.12), (3.13) and (7.2)
C_g	log-sine constant (3.14)
$\mathcal{V}_s$	geometric contrast of order s after the first return, (4.1)
$R(\tau)$	remainder of the expansion (4.14)
$\mathcal{V}_{s,U}$, U	geometric contrast over the finite range $1 \leq u \leq U$, (4.18)
ν	mean photon number of a coherent state, (4.7)
b_m, $m \geq 1$	coefficients of the compound Poisson form (4.8)
Si	sine integral, $\mathrm{Si}(x) = \int_0^x \frac{\sin y}{y}\, dy$
ξ, β	amplitude of the displacement in (4.10); complex integration variable in its proof
$\hat{\rho}$, κ, $\mathcal{L}_{\hat{O}}$	state of probe and mode, damping rate of the mode, dissipator of an operator $\hat{O}$, (5.1)
$\lvert\uparrow\rangle$, $\lvert\downarrow\rangle$	eigenstates of $\hat{\sigma}_z$ with eigenvalues 1 and -1
$\eta_n(t)$, Γ_n, $F_n(r)$	elements of the coherence of the damped mode, jump rates and gain terms, (5.2) and (5.8)
ψ_n, Δ_n	moduli $\lvert\eta_n\rvert$ and their excess $\max(\psi_n - p_n, 0)$ over p_n
$P_n(t)$, $\langle \hat{a}^\dagger \hat{a}\rangle_t$	probability of n photons and mean photon number of the damped mode
D_κ, V_κ, $\mathcal{V}^\kappa_{s,U}$	coherence, contrast and geometric contrast of the damped mode
ϵ	tolerance in Corollary 5.2
$f(t)$, T, χ_2	intrinsic coherence factor of the probe, its decay time, coefficient of a nonlinear shift
$\mathcal{P}(N)$, $u_k = N/k$, I_k	product of contrasts at harmonic delays, (6.3), the delays and the intervals between them
ζ	Riemann zeta function, $\zeta(s) = \sum_{j\geq 1} j^{-s}$ for $\sigma > 1$
$J(s)$, K_σ	integrals (7.8) and (7.7)
m, j, k, N, k_0, M	summation indices and integers
r, α, λ, δ, Y, x, y, v, w, B_1, B_2, X_1, X_2; z	auxiliary real variables and constants in the proofs; a complex variable

The function D is the coherence of the probe and V is its contrast, the quantity measured in a Ramsey experiment with delay t. The function D is the characteristic function of the geometric distribution (2.2), and the closed form below is the case without damping of the result of [6, 4]. We include its short proof.

Proposition 2.1. *For every real u,*

$$D(u) = \frac{1-q}{1-q\,e^{-2\pi iu}}, \qquad V(u) = \frac{1}{\sqrt{1+\Lambda\sin^2(\pi u)}}, \qquad \Lambda = 4\bar{n}(\bar{n}+1), \tag{2.4}$$

and

$$\log V(u) = \log p_0 - \Phi_q(u), \qquad \Phi_q(u) = \log\left|1-q\,e^{2\pi iu}\right| = -\sum_{m\geq 1}\frac{q^m}{m}\cos(2\pi mu). \tag{2.5}$$

The contrast has period 1, satisfies $1/(2\bar{n}+1) \leq V(u) \leq 1$, and $V(u) = 1$ exactly when u is an integer.

Proof. The sum in (2.3) is a geometric series of ratio $q\,e^{-2\pi iu}$, of modulus $q < 1$. This gives D. For real α, $|1-q\,e^{-i\alpha}|^2 = 1-2q\cos\alpha+q^2 = (1-q)^2+4q\sin^2(\alpha/2)$. With $\alpha = 2\pi u$ and $4q/(1-q)^2 = 4\bar{n}(\bar{n}+1)$ this gives V. Since $1-q\,e^{-2\pi iu}$ and $1-q\,e^{2\pi iu}$ are complex conjugates, $\log V(u) = \log(1-q) - \log|1-q\,e^{2\pi iu}|$. The power series of the principal logarithm, $\log(1-z) = -\sum_{m\geq 1} z^m/m$ for complex z with $|z|<1$, taken at $z = q\,e^{2\pi iu}$, has real part $-\sum_m \frac{q^m}{m}\cos(2\pi mu)$. The term $\Lambda\sin^2(\pi u)$ has period 1, vanishes exactly at the integers and is at most Λ, and $1+\Lambda = (2\bar{n}+1)^2$. □

The integers $u = 1, 2, 3, \ldots$ are the returns of the contrast. Figure 1 shows V for three temperatures.

3. The coefficients of the contrast and the transform after the first return

Throughout, $s = \sigma + i\theta$ is a complex number with real part σ and imaginary part θ. For an integer $m \geq 1$ and $\sigma > 0$ put

$$c_m(s) = \int_1^\infty \cos(2\pi mu)\,u^{-s-1}\,du. \tag{3.1}$$

Lemma 3.1. *(a) For every integer $m \geq 1$ and $\sigma > 0$,*

$$c_m(s) = \frac{s+1}{4\pi^2m^2}\Big(1-(s+2)\int_1^\infty \cos(2\pi mu)\,u^{-s-3}\,du\Big). \tag{3.2}$$

The right side is analytic for $\sigma > -2$ and is taken as the definition of $c_m(s)$ there. It satisfies

$$|c_m(s)| \leq \frac{|s+1|}{4\pi^2m^2}\Big(1+\frac{|s+2|}{\sigma+2}\Big), \qquad \sigma > -2. \tag{3.3}$$

(b) For every integer $m \geq 1$ and real $s > -1$,

$$0 < c_m(s) < \frac{s+1}{2\pi^2m^2}. \tag{3.4}$$

For real s with $-2 < s < -1$, $c_m(s) < 0$, and $c_m(-1) = 0$. (c) The coefficients at $s = 1$ satisfy

$$\sum_{m\geq 1} c_m(1) = \frac{\pi^2}{12} - \frac{3}{4}. \tag{3.5}$$

Proof. Throughout the proof m denotes an arbitrary integer with $m \geq 1$.

(a) Let $Y > 1$. Since $\sin(2\pi m) = 0$, one integration by parts gives

$$\int_1^Y \cos(2\pi mu)\,u^{-s-1}du = \frac{\sin(2\pi mY)}{2\pi m}\,Y^{-s-1} + \frac{s+1}{2\pi m}\int_1^Y \sin(2\pi mu)\,u^{-s-2}du. \tag{3.6}$$

Since $\cos(2\pi m) = 1$, a second one gives

$$\int_1^Y \sin(2\pi mu)\,u^{-s-2}du = \frac{1}{2\pi m} - \frac{\cos(2\pi mY)}{2\pi m}\,Y^{-s-2} - \frac{s+2}{2\pi m}\int_1^Y \cos(2\pi mu)\,u^{-s-3}du. \tag{3.7}$$

For $\sigma > 0$ let $Y \to \infty$ in (3.6) and (3.7). The terms with Y^{-s-1} and Y^{-s-2} tend to 0, and (3.2) follows. For $\sigma > -2$ the integral in (3.2) converges absolutely and locally uniformly in s, with modulus at most $\int_1^\infty u^{-\sigma-3}du = 1/(\sigma+2)$. Hence the right side is analytic and (3.3) holds.

(b) Let $s > -2$ be real. Then $(s+2)\int_1^\infty u^{-s-3}du = 1$. The function $\cos(2\pi mu)$ has modulus at most 1 and modulus less than 1 off a discrete set. So $(s+2)\int_1^\infty \cos(2\pi mu)u^{-s-3}du$ lies strictly between -1 and 1, and the bracket in (3.2) lies strictly between 0 and 2. The sign of $c_m(s)$ is that of $s+1$, and (3.4) follows for $s > -1$.

(c) Letting $Y \to \infty$ in (3.6) with $s = 1$ gives

$$c_m(1) = \frac{1}{\pi m}\int_1^\infty \sin(2\pi mu)\, u^{-3}\, du. \tag{3.8}$$

Let $0 < r < 1$. The series $\sum_m r^m/(\pi m)$ converges and u^{-3} is integrable on $[1,\infty)$, so the sum and the integral may be exchanged:

$$\sum_{m\geq 1} r^m\, c_m(1) = \int_1^\infty \Big(\frac{1}{\pi}\sum_{m\geq 1}\frac{r^m}{m}\sin(2\pi mu)\Big)u^{-3}\, du. \tag{3.9}$$

By the power series of the principal logarithm, $\sum_{m\geq 1}\frac{r^m}{m}\sin(2\pi mu)$ is the imaginary part of $-\log(1-re^{2\pi iu})$, that is, minus the argument of $1-re^{2\pi iu}$. This number has real part at least $1-r>0$, so its argument lies between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, and the bracket in (3.9) has modulus at most $\frac{1}{2}$. Let u be real and not an integer, and let $\alpha = 2\pi(u-\lfloor u\rfloor)$, so that $0<\alpha<2\pi$. As $r\to 1^-$, $1-re^{2\pi iu}$ tends to $1-e^{i\alpha} = 2\sin(\alpha/2)\, e^{i(\alpha/2-\pi/2)}$, with $\sin(\alpha/2)>0$ and $-\frac{\pi}{2} < \frac{\alpha}{2}-\frac{\pi}{2} < \frac{\pi}{2}$. Its argument is $\frac{\alpha}{2}-\frac{\pi}{2}$, so the bracket in (3.9) tends to $\frac{1}{2}-(u-\lfloor u\rfloor)$. By (3.4) the series $\sum_m c_m(1)$ converges absolutely, so its sum is the limit of the left side of (3.9) as $r\to 1^-$. By dominated convergence on the right side,

$$\sum_{m\geq 1} c_m(1) = \int_1^\infty \Big(\frac{1}{2}-(u-\lfloor u\rfloor)\Big)u^{-3}\, du = \sum_{N\geq 1}\int_N^{N+1}\Big(N+\frac{1}{2}-u\Big)u^{-3}\, du. \tag{3.10}$$

For an integer $N\geq 1$, $\int_N^{N+1} u^{-3}du = \frac{2N+1}{2N^2(N+1)^2}$ and $\int_N^{N+1} u^{-2}du = \frac{1}{N(N+1)}$. With $(2N+1)^2 = 4N(N+1)+1$,

$$\int_N^{N+1}\Big(N+\frac{1}{2}-u\Big)u^{-3}\, du = \frac{(2N+1)^2}{4N^2(N+1)^2} - \frac{1}{N(N+1)} = \frac{1}{4N^2(N+1)^2}. \tag{3.11}$$

Finally $\frac{1}{N^2(N+1)^2} = \big(\frac{1}{N}-\frac{1}{N+1}\big)^2 = \frac{1}{N^2}+\frac{1}{(N+1)^2}-\frac{2}{N(N+1)}$, and $\sum_{N\geq 1} N^{-2} = \frac{\pi^2}{6}$, $\sum_{N\geq 1}(N+1)^{-2} = \frac{\pi^2}{6}-1$, $\sum_{N\geq 1}\frac{1}{N(N+1)} = 1$. The sum over N of (3.11) is $\frac{1}{4}(\frac{\pi^2}{3}-3)$. □

For real u that is not an integer put $h(u) = \log|2\sin(\pi u)|$. For $\sigma>0$ and $0\leq q<1$ define

$$G(s) = \int_1^\infty h(u)\, u^{-s-1}\, du, \tag{3.12}$$

$$G_q(s) = \int_1^\infty \Phi_q(u)\, u^{-s-1}\, du, \tag{3.13}$$

with Φ_q as in (2.5). The integral (3.12) converges absolutely, because $\int_N^{N+1}|h(u)|du = \int_0^1 |h(u)|du < \infty$ for every integer N and $\sum_{N\geq 1} N^{-\sigma-1} < \infty$. The log-sine constant is

$$C_g = -\int_1^\infty \log|\sin(\pi u)|\, u^{-2}\, du = \log 2 - G(1). \tag{3.14}$$

Lemma 3.2. *For $\sigma>0$ and $0\leq q<1$,*

$$G_q(s) = -\sum_{m\geq 1}\frac{q^m}{m}\, c_m(s), \qquad G(s) = \lim_{q\to 1^-} G_q(s) = -\sum_{m\geq 1}\frac{c_m(s)}{m}. \tag{3.15}$$

The last series converges absolutely and locally uniformly for $\sigma > -2$ and continues G analytically to that half-plane. On the real axis $G(-1) = 0$, $G(s) < 0$ for $s > -1$ and $G(s) > 0$ for $-2 < s < -1$. In particular

$$C_g = \log 2 + \sum_{m\geq 1} \frac{c_m(1)}{m} = 0.743794861839400\ldots \tag{3.16}$$

Proof. For $q < 1$ the series in (2.5) is dominated by $\sum_m q^m/m$, and $u^{-\sigma-1}$ is integrable on $[1,\infty)$. Integrating term by term gives the first identity of (3.15).

For real α and $0 \leq r \leq 1$ we have $|\sin\alpha| \leq |1 - re^{i\alpha}| \leq 2$. Indeed, if $\cos\alpha \leq 0$ the modulus is at least 1, and if $\cos\alpha > 0$ it is at least the distance $|\sin\alpha|$ from the point 1 to the line through 0 and $e^{i\alpha}$. Hence $|\Phi_q(u)| \leq \log 2 + \big|\log|\sin 2\pi u|\big|$ for all q, a function of period 1 that is integrable on a period. As for (3.12), this function times $u^{-\sigma-1}$ is integrable on $[1,\infty)$. For u not an integer, $\Phi_q(u) \to \log|1 - e^{2\pi i u}| = h(u)$ as $q \to 1^-$, because $|1 - e^{i\alpha}| = 2|\sin(\alpha/2)|$. By dominated convergence $G_q(s) \to G(s)$. By (3.3) the series $\sum_m |c_m(s)|/m$ converges, locally uniformly for $\sigma > -2$, so $\sum_m q^m c_m(s)/m \to \sum_m c_m(s)/m$ as $q \to 1^-$, and the sum is analytic for $\sigma > -2$. The signs on the real axis follow from Lemma 3.1(b), and (3.16) is (3.14) with $G(1) = -\sum_m c_m(1)/m$. The decimal value is computed from this series. □

4. The sharp bound

For real $s > 0$ the weight $s\,u^{-s-1}du$ is a probability measure on $[1,\infty)$. The geometric contrast of order s after the first return is the weighted geometric mean

$$\mathcal{V}_s = \exp\Big(s\int_1^\infty \log V(u)\,u^{-s-1}\,du\Big), \qquad s > 0. \tag{4.1}$$

It depends on the temperature through $\bar n$.

Theorem 4.1. *Let $s > 0$ be real and $\bar n > 0$. Then*

$$\log \mathcal{V}_s = \log p_0 - s\,G_q(s) = \log p_0 + s\sum_{m\geq 1} \frac{q^m}{m}\,c_m(s), \qquad q = \frac{\bar n}{\bar n + 1}. \tag{4.2}$$

The ratio $\mathcal{V}_s/p_0$ is a strictly increasing function of $\bar n$. It tends to 1 as $\bar n \to 0$ and to $e^{-sG(s)}$ as $\bar n \to \infty$. Consequently

$$p_0 < \mathcal{V}_s < e^{-sG(s)}\,p_0 \qquad \text{for every } \bar n > 0, \tag{4.3}$$

and both constants are the best possible: if real numbers B_1, B_2 satisfy $B_1\,p_0 \leq \mathcal{V}_s \leq B_2\,p_0$ for every $\bar n > 0$, then $B_1 \leq 1$ and $B_2 \geq e^{-sG(s)}$. For $s = 1$,

$$p_0 < \mathcal{V}_1 < \frac{\exp(C_g)}{2}\,p_0, \qquad \frac{\exp(C_g)}{2} = 1.0519522055\ldots \tag{4.4}$$

Proof. By (2.5) and $s\int_1^\infty u^{-s-1}du = 1$, $s\int_1^\infty \log V(u)u^{-s-1}du = \log p_0 - s\,G_q(s)$, and (3.15) gives (4.2). By (3.4) every coefficient $c_m(s)$ is positive. The ratio q increases strictly from 0 to 1 as $\bar n$ increases from 0 to ∞. So $s\sum_m q^m c_m(s)/m$ is positive and strictly increasing in $\bar n$. It is at most $q\,s\sum_m c_m(s)/m$, which tends to 0 as $q \to 0$, and it tends to $-sG(s)$ as $q \to 1^-$ by (3.15). A strictly increasing function lies strictly between its limits, which is (4.3). If $B_1 p_0 \leq \mathcal{V}_s \leq B_2 p_0$ for all $\bar n$, the two limits of $\mathcal{V}_s/p_0$ give $B_1 \leq 1$ and $B_2 \geq e^{-sG(s)}$. For $s = 1$, $-G(1) = C_g - \log 2$ by (3.14). □

The bound says that the geometric contrast after the first return is the vacuum probability of the mode, up to a factor that never exceeds $e^{-sG(s)}$. The values of G on the positive real axis are therefore the logarithms of the sharp constants: $e^{-sG(s)} = 1.02005$, 1.05195, 1.14763, 1.28059 for $s = \frac12$, 1, 2, 3. Figure 2 shows G on the real axis and Figure 1 shows the ratio $\mathcal{V}_1/p_0$.

Corollary 4.2. *For every $\bar{n} > 0$,*

$$\frac{1}{\mathcal{V}_1} - 1 < \bar{n} < \frac{\exp(C_g)}{2\,\mathcal{V}_1} - 1. \tag{4.5}$$

Proof. Divide (4.4) by $\mathcal{V}_1\, p_0 > 0$ and use $1/p_0 = \bar{n} + 1$. □

The bracket (4.5) contains no fitted quantity. Its two ends differ by 5.2 per cent of $\bar{n} + 1$. For $\bar{n} = 20$, $\mathcal{V}_1 = 0.0499291$ and (4.5) reads $19.03 < \bar{n} < 20.07$.

In the ideal model the occupation is also given exactly by a single delay. By Proposition 2.1, $V(\frac{1}{2}) = 1/(2\bar{n}+1)$, which is the modulus of the mean parity $\sum_{n\geq 0}(-1)^n p_n$, so that $\bar{n} = \frac{1}{2}(1/V(\frac{1}{2}) - 1)$. Within the thermal model, therefore, the bracket (4.5) is not an improvement on the half-period contrast as an estimator of $\bar{n}$. The two contrasts measure different things, which matters when the statistics is not known (Proposition 4.4), and Theorem 4.1 fixes their relation for thermal states.

Corollary 4.3. *For every $\bar{n} > 0$,*

$$\frac{2\,V(\frac{1}{2})}{1 + V(\frac{1}{2})} < \mathcal{V}_1 < \frac{\exp(C_g)}{2} \cdot \frac{2\,V(\frac{1}{2})}{1 + V(\frac{1}{2})}. \tag{4.6}$$

Proof. By Proposition 2.1, $V(\frac{1}{2}) = 1/(2\bar{n}+1)$, so $2V(\frac{1}{2})/(1 + V(\frac{1}{2})) = 2/(2\bar{n}+2) = p_0$. Substitute this in (4.4). □

The relation (4.6) contains no parameter and holds for every thermal state, so that a pair of values that violates it is not the pair of a thermal state in the model (2.1). Other states do violate it. For a coherent state of mean photon number $\nu > 0$ the photon number has the Poisson probabilities $e^{-\nu}\nu^n/n!$, the sum in (2.3) is $\exp\big(\nu(e^{-2\pi i u} - 1)\big)$, and its modulus is $\exp\big(\nu(\cos 2\pi u - 1)\big)$. The half-period contrast is $e^{-2\nu}$, and by (3.1) the geometric contrast of order 1 after the first return is $\exp\big(-\nu(1 - c_1(1))\big)$. The quotient of the middle term of (4.6) by its left term is then

$$\frac{1 + e^{-2\nu}}{2\,e^{-2\nu}}\, e^{-\nu(1-c_1(1))} = e^{\nu\, c_1(1)} \cosh \nu. \tag{4.7}$$

It is increasing in ν. With the computed value $c_1(1) = 0.0409047$ it equals 1.0582 at $\nu = 0.3$, which is above $\frac{1}{2}\exp(C_g) = 1.0520$. Every coherent state with $\nu \geq 0.3$ therefore violates the upper inequality of (4.6).

Theorem 4.1, Corollaries 4.2 and 4.3, and the estimate of $\bar{n}$ from the half-period contrast all assume thermal statistics. We say that a probability distribution $(p_n)_{n\geq 0}$ with $p_0 > 0$ is of compound Poisson type if there are real numbers $b_m \geq 0$, $m \geq 1$, with $\sum_m b_m < \infty$, such that

$$\sum_{n\geq 0} p_n\, z^n = p_0 \exp\Big(\sum_{m\geq 1} b_m\, z^m\Big), \qquad z \in \mathbb{C}, \quad |z| \leq 1. \tag{4.8}$$

At $z = 1$ this gives $\sum_m b_m = -\log p_0$. For such a distribution we keep the definitions (2.3) of the coherence D and the contrast V, and (4.1) of the geometric contrast. The sine integral is $\mathrm{Si}(x) = \int_0^x \frac{\sin y}{y}\, dy$ [5, (6.2.9)].

Proposition 4.4. *Let $(p_n)_{n\geq 0}$ be of compound Poisson type with $p_0 < 1$.*

(a) $\log \mathcal{V}_1 = \log p_0 + \sum_{m\geq 1} b_m\, c_m(1)$, *and*

$$p_0 < \mathcal{V}_1 \leq p_0^{\,1-c_1(1)}, \qquad c_1(1) = 1 - \pi^2 + 2\pi\,\mathrm{Si}(2\pi) = 0.0409047\ldots, \tag{4.9}$$

with equality on the right exactly when $b_m = 0$ for every $m \geq 2$, that is, for the Poisson distributions. The quotient $\log \mathcal{V}_1 / \log p_0$ takes values arbitrarily close to 1 in this class, so that both exponents in (4.9) are the best possible.

(b) $\log V(\frac{1}{2}) = \log p_0 + \sum_{m\geq 1}(-1)^m b_m$, *so that $p_0^2 \leq V(\frac{1}{2}) \leq 1$. For every value of p_0 in $(0,1)$, every value in $[p_0^2, 1]$ is attained in this class.*

(c) The thermal distributions (2.2)*, the Poisson distributions and the photon-number distributions of displaced thermal states are of compound Poisson type. For the state obtained from the thermal state of ratio q, $0 < q < 1$, by a displacement of amplitude $\xi \in \mathbb{C}$,*

$$b_m = \frac{q^m}{m} + |\xi|^2(1-q)^2 q^{m-1}, \qquad p_0 = (1-q)\, e^{-|\xi|^2(1-q)}. \tag{4.10}$$

Proof. (a) At $z = e^{-2\pi i u}$, (4.8) and (2.3) give $D(u) = p_0 \exp\big(\sum_m b_m e^{-2\pi i m u}\big)$, and since the b_m are real, $\log V(u) = \log p_0 + \sum_{m\ge1} b_m \cos(2\pi m u)$, a series that converges absolutely and uniformly in u. Since $\sum_m b_m \int_1^\infty u^{-2} du < \infty$, it may be integrated term by term against u^{-2} on $[1,\infty)$, which gives the formula for $\log \mathcal{V}_1$ by (4.1) and (3.1). By (3.4) with $s = 1$, $0 < c_m(1) < 1/(\pi^2 m^2)$ for every $m \ge 1$, so that $c_m(1) < 1/(4\pi^2) = 0.0253303\ldots$ for $m \ge 2$. One integration by parts, the substitution $y = 2\pi u$ and [5, (6.2.14)] give

$$c_1(1) = \Big[-\frac{\cos 2\pi u}{u} \Big]_1^\infty - 2\pi \int_1^\infty \frac{\sin 2\pi u}{u}\, du = 1 - 2\pi\Big(\frac{\pi}{2} - \mathrm{Si}(2\pi)\Big), \tag{4.11}$$

and the computed value $\mathrm{Si}(2\pi) = 1.4181516\ldots$ gives $c_1(1) = 0.0409047\ldots > 1/(4\pi^2)$. Hence $c_m(1) \le c_1(1)$ for every m, with equality only for $m = 1$, and

$$0 < \sum_{m\ge1} b_m\, c_m(1) \le c_1(1) \sum_{m\ge1} b_m = -c_1(1) \log p_0. \tag{4.12}$$

The left inequality is strict because some b_m is positive, since $p_0 < 1$, and the right one is an equality exactly when $b_m = 0$ for $m \ge 2$. Exponentiating gives (4.9). For an integer $M \ge 1$ and $\lambda > 0$, the choice $b_M = \lambda$ and $b_m = 0$ for $m \ne M$ satisfies (4.8) with $p_0 = e^{-\lambda}$, because $e^{-\lambda}\exp(\lambda z^M)$ is the generating function of M times a Poisson variable of mean λ. It gives $\log \mathcal{V}_1 / \log p_0 = 1 - c_M(1) > 1 - 1/(\pi^2 M^2)$, which tends to 1 as $M \to \infty$.

(b) At $u = \frac12$, $\cos(\pi m) = (-1)^m$. Since $|\sum_m (-1)^m b_m| \le \sum_m b_m = -\log p_0$, we get $2\log p_0 \le \log V(\frac12) \le 0$. Let $0 \le a \le -\log p_0$, and take $b_1 = a$, $b_2 = -\log p_0 - a$ and $b_m = 0$ for $m \ge 3$. These b_m are non-negative with sum $-\log p_0$, and $p_0 \exp(b_1 z + b_2 z^2)$ is the generating function of $X_1 + 2X_2$, where X_1 and X_2 are independent Poisson variables of means b_1 and b_2; so this distribution is of compound Poisson type with vacuum probability p_0. Its half-period contrast is $p_0\, e^{-b_1+b_2} = e^{-2a}$, which takes every value in $[p_0^2, 1]$ as a runs through $[0, -\log p_0]$; $a = -\log p_0$ is the Poisson distribution.

(c) By (2.2) and the power series of $-\log(1-qz)$, $\sum_n p_n z^n = (1-q)/(1-qz) = (1-q)\exp\big(\sum_{m\ge1} q^m z^m/m\big)$. The Poisson distribution of mean ν has $\sum_n p_n z^n = e^{-\nu} e^{\nu z}$. Let $\bar n = q/(1-q)$. The thermal state of ratio q is the mixture of coherent states $|\beta\rangle$, $\beta \in \mathbb{C}$, with the Gaussian weight $(\pi\bar n)^{-1} e^{-|\beta|^2/\bar n}$ [7]. A displacement by ξ maps $|\beta\rangle$ to $|\beta+\xi\rangle$ up to a phase, so the displaced state is the mixture of the $|\beta\rangle$ with the weight $(\pi\bar n)^{-1} e^{-|\beta-\xi|^2/\bar n}$. The photon number of $|\beta\rangle$ has the generating function $e^{-|\beta|^2(1-z)}$. Hence, for $|z| \le 1$, by the Gaussian integral over β,

$$\sum_{n\ge0} p_n z^n = \frac{1}{\pi\bar n} \int_{\mathbb{C}} e^{-|\beta-\xi|^2/\bar n - (1-z)|\beta|^2}\, d^2\beta = \frac{1}{1+\bar n(1-z)} \exp\Big(-\frac{|\xi|^2(1-z)}{1+\bar n(1-z)} \Big). \tag{4.13}$$

Now $1 + \bar n(1-z) = (\bar n + 1)(1-qz)$ and $(1-z)/(1-qz) = 1 - (1-q)\sum_{m\ge1} q^{m-1} z^m$. With $1/(\bar n + 1) = 1 - q$, the right side of (4.13) is $(1-q) e^{-|\xi|^2(1-q)} \exp\big(\sum_m b_m z^m\big)$ with the b_m of (4.10), which are non-negative. □

Table 2 illustrates Proposition 4.4 with the Poisson distribution of mean 1 and with displaced thermal states of thermal occupation $\frac12$, that is, modes that hold thermal photons and a coherent field at the same time. In every row $\log \mathcal{V}_1 / \log p_0$ lies between $1 - c_1(1) = 0.9591$ and 1, as (4.9) requires. The last column is the vacuum probability that the half-period contrast gives under the thermal hypothesis, $2V(\frac12)/(1+V(\frac12))$, divided by the true p_0. It is exact for the thermal state, 9 per cent too low for $|\xi|^2 = 1$ and 35 per cent too low for the Poisson distribution.

TABLE 2. Vacuum probability p_0, geometric contrast $\mathcal{V}_1$ and half-period contrast $V(\frac{1}{2})$ for the Poisson distribution of mean 1 and for the displaced thermal states (4.10) with $q = \frac{1}{3}$ (thermal occupation $\frac{1}{2}$). The last column is $2V(\frac{1}{2})/(1+V(\frac{1}{2}))$, the vacuum probability given by the half-period contrast under the thermal hypothesis, divided by p_0.

state	p_0	$\mathcal{V}_1$	$\log \mathcal{V}_1 / \log p_0$	$V(\frac{1}{2})$	last column
Poisson, mean 1	0.36788	0.38324	0.9591	0.13534	0.648
$q = \frac{1}{3}$, $\lvert\xi\rvert^2 = 0$	0.66667	0.67632	0.9646	0.50000	1.000
$q = \frac{1}{3}$, $\lvert\xi\rvert^2 = 0.25$	0.56432	0.57540	0.9660	0.38940	0.993
$q = \frac{1}{3}$, $\lvert\xi\rvert^2 = 1$	0.34228	0.35434	0.9677	0.18394	0.908
$q = \frac{1}{3}$, $\lvert\xi\rvert^2 = 2$	0.17573	0.18565	0.9684	0.06768	0.721

This is the comparative statement that the note proves. With an unknown mixture of thermal and coherent photons, the geometric contrast after the first return determines the vacuum probability within the factor $p_0^{-c_1(1)}$: the ratio $\log \mathcal{V}_1 / \log p_0$ lies in $[1 - c_1(1), 1)$. This is an exact property of the ideal contrast, not an experimental error bar. The half-period contrast is the modulus of the mean parity $D(\frac{1}{2}) = \sum_n (-1)^n p_n$ and gives p_0 only through a hypothesis on the statistics. The vacuum probability is the quantity read by a pulse selective on the zero-photon line [9, 15]; the geometric contrast reaches it without selective pulses. Within the thermal model the half-period contrast remains the simpler way to obtain $\bar{n}$, and Section 5 shows that it is also less affected by damping.

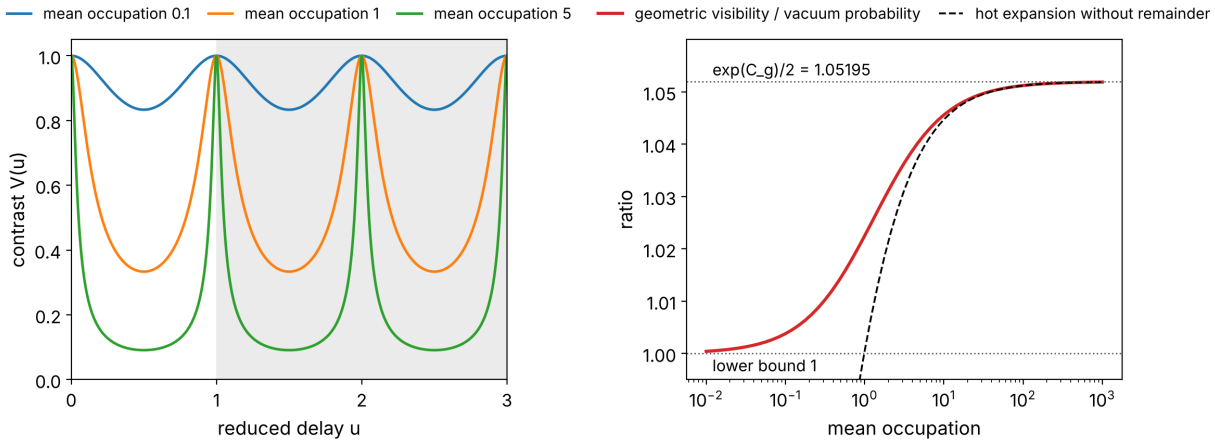


FIGURE 1. Left: the contrast $V(u)$ of (2.4) for $\bar{n} = 0.1$, 1 and 5; the shaded region $u \geq 1$ is the range of the geometric contrast (4.1). Right: the ratio $\mathcal{V}_1/p_0$ as a function of $\bar{n}$, between its sharp bounds 1 and $\exp(C_g)/2$ of (4.4), and the expansion (4.14) without its remainder.

At high temperature the ratio approaches its upper bound, and the approach can be made explicit.

Theorem 4.5. *Let $\bar{n} > 0$ and $\tau = \log(1/q) = \log(1 + 1/\bar{n})$. Then*

$$\log \mathcal{V}_1 = \log p_0 + C_g - \log 2 - \Big(\frac{\pi^2}{12} - \frac{3}{4}\Big)\tau + R(\tau), \qquad R(\tau) > 0, \tag{4.14}$$

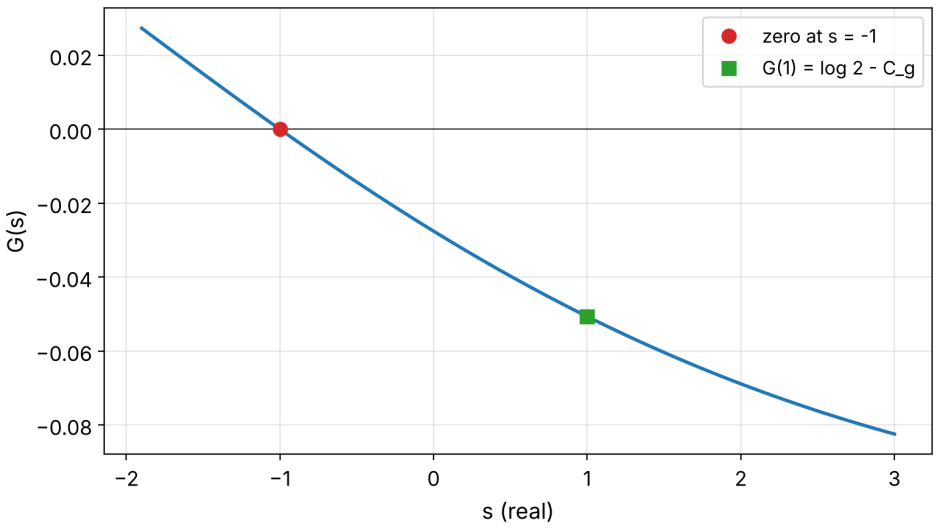


FIGURE 2. The function G on the real axis. For $s > 0$ the sharp constant of (4.3) is $e^{-sG(s)}$; the square marks $G(1) = \log 2 - C_g$.

that is, $\mathcal{V}_1 = \frac{1}{2}\exp(C_g)\,p_0\,q^{\pi^2/12-3/4}\,e^{R(\tau)}$. *For* $0 < \tau \le 1$, *that is, for* $\bar n \ge 1/(e-1)$,

$$R(\tau) \le \frac{\tau^2}{\pi^2}\Big(\frac{5}{2} + \frac{1}{2}\log\frac{1}{\tau}\Big). \tag{4.15}$$

Proof. Write $c_m = c_m(1)$ and $q^m = e^{-m\tau}$. By (4.2), (3.16) and (3.5),

$$R(\tau) = \sum_{m\ge1}\frac{q^m}{m}c_m - \sum_{m\ge1}\frac{c_m}{m} + \tau\sum_{m\ge1}c_m = \sum_{m\ge1}\frac{c_m}{m}(m\tau - 1 + e^{-m\tau}). \tag{4.16}$$

For real $y > 0$, $0 < y - 1 + e^{-y} \le \min(y, \frac{1}{2}y^2)$. Since $c_m > 0$, this gives $R(\tau) > 0$. With $c_m < 1/(\pi^2 m^2)$ from (3.4) it also gives

$$R(\tau) \le \frac{1}{\pi^2}\sum_{m\ge1}\min\Big(\frac{\tau}{m^2}, \frac{\tau^2}{2m}\Big). \tag{4.17}$$

Let $0 < \tau \le 1$ and let $M = \lfloor 1/\tau \rfloor \ge 1$. Then $\sum_{m\le M}\frac{1}{m} \le 1 + \log M \le 1 + \log\frac{1}{\tau}$ and $\sum_{m>M}\frac{1}{m^2} < \frac{1}{M} \le 2\tau$, because $\lfloor y\rfloor \ge \frac{y}{2}$ for $y \ge 1$. Using the second entry of the minimum for $m \le M$ and the first for $m > M$, the sum in (4.17) is at most $\frac{\tau^2}{2}(1 + \log\frac{1}{\tau}) + 2\tau^2$. □

Table 3 gives the exact value of $\mathcal{V}_1$ from (4.2), the ratio $\mathcal{V}_1/p_0$, the remainder $R(\tau)$ and its bound (4.15). The bound (4.15) is not sharp: in the table the remainder is smaller than the bound by a factor between 2.9 and 6.

A measured sequence of contrasts covers a finite range of delays. For real $U > 1$ and $s > 0$ let

$$\mathcal{V}_{s,U} = \exp\Big(\frac{s}{1 - U^{-s}}\int_1^U \log V(u)\,u^{-s-1}\,du\Big), \tag{4.18}$$

the geometric mean of the contrast over $1 \le u \le U$ with the normalised weight.

Proposition 4.6. *Let* $\bar n > 0$, $L = \log(2\bar n + 1)$, $s > 0$ *and* $U > 1$. *Then*

$$\big|\log\mathcal{V}_{s,U} - \log\mathcal{V}_s\big| \le L\,U^{-s}. \tag{4.19}$$

TABLE 3. The geometric contrast $\mathcal{V}_1$ computed from (4.2), its ratio to the vacuum probability p_0, the remainder $R(\tau)$ of (4.14) and the bound (4.15), which applies when $\tau \le 1$.

$\bar n$	p_0	$\mathcal{V}_1$	$\mathcal{V}_1/p_0$	$R(\tau)$	bound (4.15)
0.01	0.9900990	0.9905007	1.0004057	2.84×10^{-1}	
0.1	0.9090909	0.9125237	1.0037761	1.27×10^{-1}	
1	0.5000000	0.5112376	1.0224752	2.18×10^{-2}	1.31×10^{-1}
5	0.1666667	0.1734625	1.0407747	2.53×10^{-3}	1.13×10^{-2}
20	0.0476190	0.0499291	1.0485104	2.58×10^{-4}	9.67×10^{-4}
100	0.0099010	0.0104080	1.0512094	1.47×10^{-5}	4.82×10^{-5}
1000	0.0009990	0.0010508	1.0518762	2.07×10^{-7}	6.03×10^{-7}

Proof. Put $X_1 = s\int_1^U \log V(u)\, u^{-s-1}du$ and $X_2 = s\int_U^\infty \log V(u)\, u^{-s-1}du$, so that $\log \mathcal{V}_s = X_1 + X_2$ and $\log \mathcal{V}_{s,U} = X_1/(1-U^{-s})$. By Proposition 2.1, $-L \le \log V(u) \le 0$. Since $s\int_1^U u^{-s-1}du = 1-U^{-s}$ and $s\int_U^\infty u^{-s-1}du = U^{-s}$, we have $-L(1-U^{-s}) \le X_1 \le 0$ and $-L\,U^{-s} \le X_2 \le 0$. The difference $\log \mathcal{V}_{s,U} - \log \mathcal{V}_s = X_1U^{-s}/(1-U^{-s}) - X_2$ is thus the sum of a number in $[-L\,U^{-s}, 0]$ and a number in $[0, L\,U^{-s}]$. □

The bound (4.19) uses only the range of $\log V$ and is not sharp: for $\bar n = 20$, $s = 1$ and $U = 20$ it gives 0.186, while the computed difference is 0.0025.

5. Damping, intrinsic dephasing and a nonlinear shift

The returns of the contrast to 1 rest on three idealisations: no damping of the mode, no intrinsic dephasing of the probe, and a shift exactly linear in the photon number. We bound the effect of the first and discuss the other two.

Let the mode be damped at the rate $\kappa > 0$ by a bath with the same occupation $\bar n$, so that the thermal state of the mode is stationary. In the Lindblad description used for this problem [12, 4, 15], the state $\hat\rho$ of probe and mode obeys

$$\frac{d\hat\rho}{dt} = -i[\hat H, \hat\rho] + \kappa(\bar n+1)\,\mathcal{L}_{\hat a}\hat\rho + \kappa\bar n\,\mathcal{L}_{\hat a^\dagger}\hat\rho, \qquad \mathcal{L}_{\hat O}\hat\rho = \hat O\hat\rho\hat O^\dagger - \frac{1}{2}(\hat O^\dagger\hat O\hat\rho + \hat\rho\hat O^\dagger\hat O), \tag{5.1}$$

with $\hat H$ as in (2.1). Initially the probe is in $(|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$, where $\hat\sigma_z|\uparrow\rangle = |\uparrow\rangle$ and $\hat\sigma_z|\downarrow\rangle = -|\downarrow\rangle$, and the mode is thermal. For $n \ge 0$ let $\eta_n(t)$ be $2e^{i\omega_p t}$ times the diagonal element in the state with n photons of the operator $\langle\uparrow|\hat\rho(t)|\downarrow\rangle$ on the mode, so that $\eta_n(0) = p_n$. The coherence of the damped mode is $D_\kappa(t) = \sum_n \eta_n(t)$ and its contrast is $V_\kappa(t) = |D_\kappa(t)|$; without damping $D_\kappa(t) = D(\chi t/\pi)$. The operator $\langle\uparrow|\hat\rho|\downarrow\rangle$ starts diagonal in the photon number, and the Hamiltonian part and both dissipators of (5.1) map such operators to operators of the same kind. With $\eta_{-1} = 0$ and

$$\frac{d\eta_n}{dt} = -(2i\chi n + \Gamma_n)\,\eta_n + \kappa(\bar n+1)(n+1)\,\eta_{n+1} + \kappa\bar n\,n\,\eta_{n-1}, \qquad \Gamma_n = \kappa\big[(\bar n+1)n + \bar n(n+1)\big], \tag{5.2}$$

these are the rate equations of [15] with the phase of each photon number added, the system solved in closed form in [6, 4]. The thermal distribution satisfies detailed balance: since $p_{n+1} = qp_n$ and $q(\bar n+1) = \bar n$,

$$\kappa(\bar n+1)(n+1)\,p_{n+1} + \kappa\bar n\,n\,p_{n-1} = \Gamma_n\,p_n, \qquad n \ge 0. \tag{5.3}$$

Proposition 5.1. *Let (η_n) be the solution of (5.2) with $\eta_n(0) = p_n$ for which $\sum_n (n+1)|\eta_n(t)|$ is bounded on bounded intervals of t. The solution given by the density operator of (5.1) has this property, with the bound $\bar n + 1$ (Step 0 of the proof). Then, for every $t \ge 0$,*

$$|D_\kappa(t) - D(\chi t/\pi)| \le 2\sum_{n\ge0} p_n(1 - e^{-\Gamma_n t}) \le \Lambda\,\kappa\,t, \qquad \Lambda = 4\bar n(\bar n+1), \tag{5.4}$$

and in particular $|V_\kappa(t) - V(u)| \le \pi\Lambda\,(\kappa/\chi)\,u$ *with* $u = \chi t/\pi$.

Proof. Step 0. Let $P_n(t)$ be the probability of n photons in $\hat\rho(t)$. Positivity of $\hat\rho(t)$ gives $|\langle n,\uparrow|\hat\rho|n,\downarrow\rangle|^2 \le \langle n,\uparrow|\hat\rho|n,\uparrow\rangle\langle n,\downarrow|\hat\rho|n,\downarrow\rangle$, so that $|\eta_n(t)| \le P_n(t)$ by the inequality between the geometric and the arithmetic mean. The Hamiltonian $\hat H$ commutes with $\hat a^\dagger\hat a$, and $\hat a^\dagger(\hat a^\dagger\hat a)\hat a - \frac12\{\hat a^\dagger\hat a, \hat a^\dagger\hat a\} = -\hat a^\dagger\hat a$, $\hat a(\hat a^\dagger\hat a)\hat a^\dagger - \frac12\{\hat a\hat a^\dagger, \hat a^\dagger\hat a\} = \hat a^\dagger\hat a + 1$. Hence the mean photon number $\langle\hat a^\dagger\hat a\rangle_t = \sum_n nP_n(t)$ obeys

$$\frac{d\langle\hat a^\dagger\hat a\rangle_t}{dt} = -\kappa(\bar n+1)\langle\hat a^\dagger\hat a\rangle_t + \kappa\bar n\big(\langle\hat a^\dagger\hat a\rangle_t + 1\big) = -\kappa\big(\langle\hat a^\dagger\hat a\rangle_t - \bar n\big), \tag{5.5}$$

so that $\langle\hat a^\dagger\hat a\rangle_t = \bar n$ for all $t \ge 0$, and $\sum_n(n+1)|\eta_n(t)| \le \sum_n(n+1)P_n(t) = \bar n + 1$.

Step 1: $|\eta_n(t)| \le p_n$ for all n and $t \ge 0$. Let $\psi_n = |\eta_n|$ and $\Delta_n = \max(\psi_n - p_n, 0)$. Since $\mathrm{Re}\,(\bar\eta_n \cdot (-2i\chi n)\eta_n) = 0$, (5.2) gives, for $t_1 < t_2$,

$$\psi_n(t_2) - \psi_n(t_1) \le \int_{t_1}^{t_2} \Big(-\Gamma_n\psi_n + \kappa(\bar n+1)(n+1)\psi_{n+1} + \kappa\bar n\, n\,\psi_{n-1}\Big)dr. \tag{5.6}$$

Subtracting (5.3) and using $\psi_k - p_k \le \Delta_k$, the same inequality holds for Δ_n on every interval on which $\psi_n > p_n$, with Δ in place of ψ. Since Δ_n is absolutely continuous, vanishes at $t = 0$ and is constant on the intervals where it is 0, this gives $\Delta_n(t) \le \int_0^t \big(-\Gamma_n\Delta_n + \kappa(\bar n+1)(n+1)\Delta_{n+1} + \kappa\bar n\, n\,\Delta_{n-1}\big)dr$. Let $N \ge 0$ be an integer and sum these inequalities over $0 \le n \le N$. Since $\Gamma_k = \kappa(\bar n+1)k + \kappa\bar n(k+1)$, the gain terms with index $k \le N$ cancel the loss terms, and what remains is

$$\sum_{n=0}^{N}\Delta_n(t) \le \kappa(N+1)\int_0^t \Big((\bar n+1)\Delta_{N+1}(r) - \bar n\,\Delta_N(r)\Big)dr \le \kappa(\bar n+1)\int_0^t (N+1)\,\psi_{N+1}(r)\,dr. \tag{5.7}$$

By hypothesis $\sum_n(n+1)\psi_n(r)$ is bounded for $0 \le r \le t$. Hence $(N+1)\psi_{N+1}(r)$ is bounded uniformly in N and r, and it tends to 0 as $N \to \infty$ for each r, as the terms of a convergent series do. By dominated convergence the right side of (5.7) tends to 0. The left side is non-negative and non-decreasing in N, so $\sum_{n\ge0}\Delta_n(t) = 0$, and every $\Delta_n(t)$ is 0.

Step 2. Put $F_n(r) = \kappa(\bar n+1)(n+1)\eta_{n+1}(r) + \kappa\bar n\, n\,\eta_{n-1}(r)$. Solving (5.2) as a linear equation for η_n,

$$\eta_n(t) = p_n\, e^{-(2i\chi n+\Gamma_n)t} + \int_0^t e^{-(2i\chi n+\Gamma_n)(t-r)}\,F_n(r)\,dr. \tag{5.8}$$

By Step 1 and (5.3), $|F_n(r)| \le \Gamma_n p_n$. Hence

$$\big|\eta_n(t) - p_n\,e^{-2i\chi nt}\big| \le p_n(1 - e^{-\Gamma_n t}) + \int_0^t e^{-\Gamma_n(t-r)}\,\Gamma_n p_n\,dr = 2p_n(1 - e^{-\Gamma_n t}). \tag{5.9}$$

Summing over n, with $D(\chi t/\pi) = \sum_n p_n e^{-2i\chi nt}$, gives the first inequality of (5.4). Since $1 - e^{-y} \le y$ for $y \ge 0$ and $\sum_n np_n = \bar n$, the sum is at most $2t\sum_n p_n\Gamma_n = 2\kappa t\big((\bar n+1)\bar n + \bar n(\bar n+1)\big) = \Lambda\kappa t$. Finally $||a| - |b|| \le |a-b|$ and $t = \pi u/\chi$. □

For the geometric contrast of the damped mode over $1 \le u \le U$, $\mathcal{V}^\kappa_{s,U}$, defined by (4.18) with $V_\kappa(\pi u/\chi)$ in place of $V(u)$, the bound gives the following.

Corollary 5.2. *Let* $s > 0$, $U > 1$, *and suppose that* $\epsilon = \pi\Lambda(2\bar n+1)(\kappa/\chi)\,U < 1$. *Then*

$$\big|\log\mathcal{V}^\kappa_{s,U} - \log\mathcal{V}_{s,U}\big| \le -\log(1-\epsilon), \qquad \Big|\log V_\kappa\Big(\frac{\pi}{2\chi}\Big) - \log V\Big(\frac12\Big)\Big| \le -\log\Big(1 - \frac{\epsilon}{2U}\Big). \tag{5.10}$$

Proof. For $0 \le u \le U$, Proposition 5.1 and $V(u) \ge 1/(2\bar n+1)$ give $|V_\kappa(\pi u/\chi) - V(u)| \le \pi\Lambda(\kappa/\chi)U = \epsilon/(2\bar n+1) \le \epsilon V(u)$. So $1-\epsilon \le V_\kappa/V \le 1+\epsilon$ and $|\log(V_\kappa/V)| \le -\log(1-\epsilon)$, because $\log(1+\epsilon) \le -\log(1-\epsilon)$. The weight in (4.18) is a probability measure on $[1,U]$. At $u = \frac12$, Proposition 5.1 gives $|V_\kappa(\frac{\pi}{2\chi}) - V(\frac12)| \le \pi\Lambda(\kappa/\chi)/2 = \epsilon/(2U(2\bar n+1)) \le \frac{\epsilon}{2U}V(\frac12)$, and the logarithm is bounded as before with $\frac{\epsilon}{2U}$ in place of ϵ. □

The condition $\epsilon < 1$ asks for $\kappa/\chi < 1/\big(\pi\Lambda(2\bar{n}+1)U\big)$, which decreases as $\bar{n}^{-3}$. With $U = 20$ and $\epsilon = 0.1$ it reads $\kappa/\chi < 3.0 \times 10^{-3}$ for $\bar{n} = 0.1$, $\kappa/\chi < 6.6 \times 10^{-5}$ for $\bar{n} = 1$ and $\kappa/\chi < 2.3 \times 10^{-8}$ for $\bar{n} = 20$. For $\chi/2\pi = 1$ MHz these are $\kappa/2\pi < 3.0$ kHz, 66 Hz and 0.023 Hz. Cavity linewidths $\kappa/2\pi$ between 6 and 120 kHz are reported in [15], and a single-photon lifetime of 25.6 ms, that is $\kappa/2\pi = 6.2$ Hz, in [10]. The geometric contrast is therefore usable at low occupation in long-lived modes, and the statements of Section 4 at high occupation, such as the example $\bar{n} = 20$ after Corollary 4.2, are statements about the ideal model only. The half-period contrast is affected less, by the factor $1/(2U)$ in (5.10), because its delay is short. The bound (5.4) is not sharp: for $\bar{n}$ in $\{0.05, 1, 5, 20\}$, $\kappa/(2\chi)$ between 10^{-5} and 10^{-2} and $0 < u \le 6$, the difference $|V_\kappa - V|$ computed from the closed form of [4] is at most 0.51 times the bound.

Intrinsic dephasing of the probe. If, independently of the photons, the probe has a coherence factor $f(t)$ with $0 < f(t) \le 1$, the measured contrast without damping is $f(t)V(u)$. Since $V(k) = 1$ at the returns, the contrast measured at $u = k$ is $f(\pi k/\chi)$: the returns measure the decay of the probe itself, which can then be divided out. Without that correction, a factor $f(t) = e^{-t/T}$ changes $\log \mathcal{V}_{1,U}$ by

$$\frac{1}{1-U^{-1}} \int_1^U \Big(-\frac{\pi u}{\chi T}\Big) u^{-2}\, du = -\frac{\pi}{\chi T} \cdot \frac{U \log U}{U-1}. \tag{5.11}$$

Nonlinear shift. If the phase acquired by the component with n photons is not exactly $2\chi n t$, for example $2\chi n t + \chi_2 n(n-1)t$ with a real $\chi_2 \neq 0$, the contrast at the return $u = k$ becomes $\big|\sum_n p_n e^{-i\pi k \chi_2 n(n-1)/\chi}\big|$, which is below 1 in general. With the decay of the probe divided out, a measured contrast below 1 at the returns therefore signals damping or a nonlinear shift, and Proposition 5.1 bounds the part due to damping. None of the results of Sections 4, 6 and 7 is claimed for a nonlinear shift.

6. The sum rule and the harmonic delays

The first return divides the delays into the interval $0 < u < 1$ and the half-line $u \ge 1$. Over all delays together the integral of the logarithm of the contrast is linear in the occupation.

Proposition 6.1. *For every $\bar{n} > 0$,*

$$\int_0^\infty \log V(u)\, u^{-2}\, du = -\pi^2\, \bar{n}, \qquad \int_0^1 \log V(u)\, u^{-2}\, du = -\pi^2\, \bar{n} - \log \mathcal{V}_1. \tag{6.1}$$

Proof. By (2.5) and $\log(1-q) = -\sum_m q^m/m$, $\log V(u) = \sum_{m\ge1} \frac{q^m}{m}\big(\cos(2\pi m u) - 1\big)$, a series of terms that are all ≤ 0, so by Tonelli's theorem it may be integrated term by term. Let $\lambda > 0$ and $0 < \delta < Y$. Integration by parts gives $\int_\delta^Y (1-\cos\lambda u)u^{-2}du = \frac{1-\cos\lambda\delta}{\delta} - \frac{1-\cos\lambda Y}{Y} + \lambda\int_\delta^Y \frac{\sin\lambda u}{u}\, du$. As $\delta \to 0$ and $Y \to \infty$ the first two terms tend to 0 and the last integral tends to $\frac{\pi}{2}$, by the classical Dirichlet integral [5, (6.2.9), (6.2.14)]. Hence $\int_0^\infty (1-\cos\lambda u)u^{-2}du = \frac{\pi\lambda}{2}$. With $\lambda = 2\pi m$,

$$\int_0^\infty \log V(u)\, u^{-2}\, du = -\sum_{m\ge1} \frac{q^m}{m}\, \pi^2 m = -\pi^2\, \frac{q}{1-q} = -\pi^2\, \bar{n}. \tag{6.2}$$

The second identity of (6.1) follows from $\int_1^\infty \log V(u) u^{-2} du = \log \mathcal{V}_1$, which is (4.1) with $s = 1$. □

By (4.4) the part of the sum rule that lies after the first return, $\log \mathcal{V}_1$, differs from $\log p_0$ by less than $C_g - \log 2 = 0.0506\ldots$ for every positive occupation. The part before the first return carries the linear term $-\pi^2\bar{n}$.

The logarithm of the geometric contrast is also the logarithmic rate of decay of a product. For an integer $N \ge 2$ let

$$\mathcal{P}(N) = \prod_{k=1}^{N-1} V\Big(\frac{N}{k}\Big). \tag{6.3}$$

This is the product of the contrasts measured at the harmonic delays $u_k = N/k$, that is, at the times $N\pi/(k\chi)$. It is also the contrast at the time $N\pi/\chi$ of a probe coupled to $N-1$ independent modes with dispersive couplings χ/k, $1 \le k \le N-1$, and equal occupation $\bar n$, because the coherences of independent modes multiply.

Proposition 6.2. *Let $\bar n > 0$, $L = \log(2\bar n + 1)$ and let $N \ge 2$ be an integer. Then*

$$\Big|\frac{1}{N}\log \mathcal{P}(N) - \log \mathcal{V}_1\Big| \le L\Big(\frac{2\sqrt{2}}{\sqrt{N}} + \frac{1}{N}\Big). \tag{6.4}$$

In particular $\mathcal{P}(N)^{1/N} \to \mathcal{V}_1$ as $N \to \infty$, and by (4.4) this limit lies strictly between p_0 and $\frac{1}{2}\exp(C_g)p_0$.

Proof. By Proposition 2.1 the function $\log V$ has period 1 and takes values in $[-L, 0]$. On each interval $[j, j+\frac12]$ with integer j it decreases from 0 to $-L$, and on $[j+\frac12, j+1]$ it increases from $-L$ to 0. Put $u_k = N/k$ for $1 \le k \le N$, so that $u_N = 1$ and $u_1 = N$, and $I_k = [u_{k+1}, u_k]$ for $1 \le k \le N-1$. The intervals I_k cover $[1, N]$ and overlap only at end points, and $\int_{I_k} u^{-2}du = \frac{k+1}{N} - \frac{k}{N} = \frac{1}{N}$. Therefore

$$\frac{1}{N}\log \mathcal{P}(N) - \int_1^N \log V(u)\, u^{-2}\, du = \sum_{k=1}^{N-1} \int_{I_k} \big(\log V(u_k) - \log V(u)\big)\, u^{-2}\, du. \tag{6.5}$$

The k-th term has modulus at most $\frac{1}{N}$ times the oscillation of $\log V$ on I_k, which is at most L and at most the total variation of $\log V$ on I_k. Let k_0 be an integer with $1 \le k_0 \le N-1$. The intervals I_k with $k > k_0$ cover $[1, u_{k_0+1}]$ and overlap only at end points, so the sum of the total variations on them is the total variation on $[1, u_{k_0+1}]$. An interval $[1, x]$ with $x \ge 1$ is covered by the intervals of monotonicity $[\frac{j}{2}, \frac{j+1}{2}]$ with $2 \le j \le \lceil 2x\rceil - 1$, at most $2x-1$ in number, on each of which the variation is at most L. So the total variation of $\log V$ on $[1, x]$ is at most $2Lx$. The right side of (6.5) is thus at most $\frac{L}{N}(k_0 + \frac{2N}{k_0+1})$ in modulus. Choose $k_0 = \lceil\sqrt{2N}\rceil - 1$. Since $N \ge 2$, $2 \le \sqrt{2N} \le N$, so $1 \le k_0 \le N-1$, and $k_0 < \sqrt{2N} \le k_0 + 1$. Then $k_0 + \frac{2N}{k_0+1} < 2\sqrt{2N}$, which gives the first term of (6.4). Finally $\big|\int_N^\infty \log V(u)u^{-2}du\big| \le \frac{L}{N}$, and $\int_1^\infty \log V(u)u^{-2}du = \log \mathcal{V}_1$. □

At zero temperature the factors of the same product, formed with $2|\sin(\pi N/k)|$ in place of the contrast, vanish whenever k divides N. The thermal factors do not vanish: by Proposition 2.1 each of them is at least $1/(2\bar n + 1)$, and a divisor k of N contributes the factor 1.

7. The transforms after and before the first return and their zeros

Theorem 4.1 uses G on the positive real axis. At complex exponents the same function describes the hot limit of the contrast against weights that oscillate in the logarithm of the delay.

Proposition 7.1. *For $\sigma > 0$,*

$$\lim_{\bar n \to \infty} \int_1^\infty \log\frac{V(u)}{p_0}\, u^{-s-1}\, du = -G(s), \tag{7.1}$$

uniformly for s in compact subsets of the half-plane $\sigma > 0$.

Proof. By (2.5) the integral is $-G_q(s)$. By (3.15), $|G_q(s) - G(s)| \le \sum_{m\ge1}(1-q^m)|c_m(s)|/m$. By (3.3) the series $\sum_m |c_m(s)|/m$ converges uniformly on compact subsets of $\sigma > -2$, and each term of the bound tends to 0 as $q \to 1^-$, so the bound tends to 0 uniformly there. □

For real $s > 0$ the limit (7.1) is positive and gives the sharp constants of Theorem 4.1. For $s = \sigma + i\theta$ with $\theta \ne 0$ the weight is $u^{-\sigma-1}e^{-i\theta\log u}$. At a zero of G with $\sigma > 0$ the limit vanishes: the excess $\log(V/p_0)$ of a hot mode has no component on that weight. Table 4 lists the zeros of G computed in the rectangle $0 < \sigma < 2$, $0 < \theta \le 40$, and Figure 3 shows $|G|$ there. The zeros were computed numerically. Local minima of $\log|G|$ on a grid of step 0.05 were refined by

Newton's method in double precision and then with 20-digit arithmetic, which leaves $|G| < 10^{-20}$ at the listed points. The argument principle, evaluated numerically on the boundary of the larger rectangle $-1.5 < \sigma < 3$, $0.2 < \theta < 200$, gives 79, the number of zeros found there. This is a numerical check; we do not claim that the list is complete. The real zeros of A in Table 4 were computed with a root finder on the real axis in 25-digit arithmetic, with $|A| < 10^{-25}$ at the listed points.

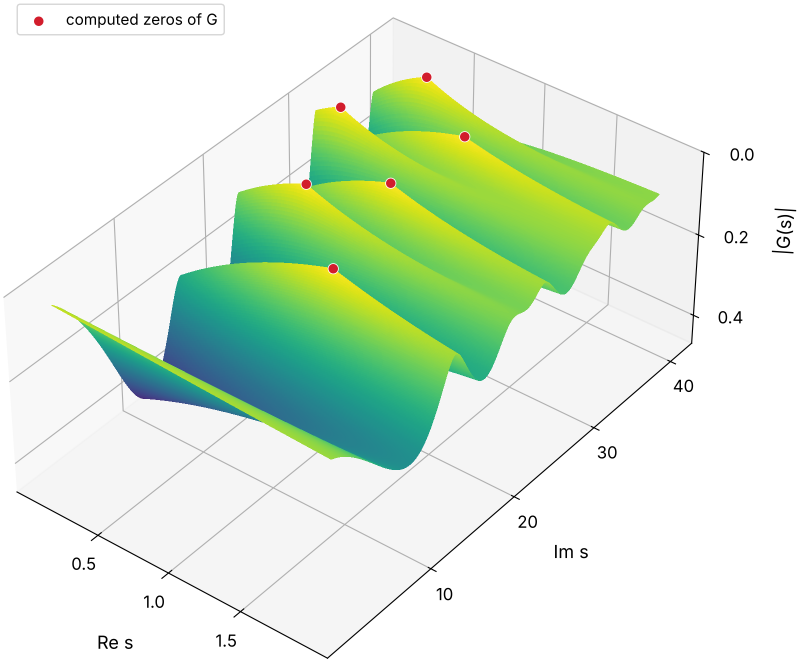


FIGURE 3. $|G(s)|$ on the rectangle $0.05 \le \operatorname{Re} s \le 1.95$, $1 \le \operatorname{Im} s \le 40$, on a linear vertical axis that is inverted, so that a zero is a peak. The red markers are the six computed zeros of G of Table 4.

The delays before the first return enter through a second function. The weight u^{-s-1} is not integrable at $u = 0$ against h, which behaves there as $\log(2\pi u)$. Subtracting this short-time form gives, for $\sigma < 2$ and $s \neq 0$,

$$A(s) = \frac{\log(2\pi)}{s} + \frac{1}{s^2} - \int_0^1 \log \frac{\sin(\pi u)}{\pi u}\, u^{-s-1}\, du. \tag{7.2}$$

The integral converges absolutely for $\sigma < 2$, because the logarithm is $O(u^2)$ as $u \to 0$ and has an integrable singularity at $u = 1$. Let ζ denote the Riemann zeta function, $\zeta(s) = \sum_{j \ge 1} j^{-s}$ for $\sigma > 1$, continued analytically to the complex plane with a simple pole at $s = 1$.

Proposition 7.2. *(a) For $\sigma < 2$ and $s \neq 0$,*

$$A(s) = \frac{\log(2\pi)}{s} + \frac{1}{s^2} + \sum_{k \ge 1} \frac{\zeta(2k)}{k\,(2k - s)}. \tag{7.3}$$

The series converges for every complex s other than $2, 4, 6, \ldots$ and continues A to a meromorphic function on the complex plane.

(b) For $0 < \sigma < 2$,

$$G(s) - A(s) = \frac{\pi}{s} \cot\Big(\frac{\pi s}{2}\Big)\, \zeta(s), \tag{7.4}$$

where the right side is continued analytically to $s = 1$, with value $-\frac{\pi^2}{2}$.

(c) For non-real s with $0 < \sigma < 2$, $G(s) = A(s)$ if and only if $\zeta(s) = 0$.

Proof. (a) By the product for the sine [5, (4.22.1)], $\sin(\pi u) = \pi u \prod_{j\ge1}(1 - u^2/j^2)$. For $0 < u < 1$ every factor lies in $(0, 1)$ and

$$\log \frac{\sin(\pi u)}{\pi u} = \sum_{j\ge1} \log\Big(1 - \frac{u^2}{j^2}\Big) = -\sum_{j\ge1}\sum_{k\ge1} \frac{u^{2k}}{k\, j^{2k}} = -\sum_{k\ge1} \frac{\zeta(2k)}{k}\, u^{2k}, \tag{7.5}$$

a double series of terms of one sign. For $\sigma < 2$, $\int_0^1 u^{2k-s-1} du = \frac{1}{2k-s}$, and $\sum_k \frac{\zeta(2k)}{k(2k-\sigma)} < \infty$. Integrating (7.5) term by term in (7.2) gives (7.3). Since $\zeta(2k) \le \zeta(2)$, the terms are $O(k^{-2})$ uniformly on compact sets that avoid the points $2k$, which gives the continuation.

(b) For $\sigma > 0$, $\int_1^\infty \log(2\pi u)\, u^{-s-1} du = \frac{\log(2\pi)}{s} + \frac{1}{s^2}$. For $0 < u < 1$, $h(u) - \log(2\pi u) = \log \frac{\sin(\pi u)}{\pi u}$. Hence, for $0 < \sigma < 2$,

$$G(s) - A(s) = \int_0^\infty \big(h(u) - \log(2\pi u)\big)\, u^{-s-1}\, du. \tag{7.6}$$

By the product for the sine, $h(u) - \log(2\pi u) = \sum_{j\ge1} \log|1 - u^2/j^2|$ for every $u > 0$ that is not an integer. For $0 < \sigma < 2$ put

$$K_\sigma = \int_0^\infty \big|\log|1 - v^2|\big|\, v^{-\sigma-1}\, dv < \infty. \tag{7.7}$$

It is finite because the logarithm is $O(v^2)$ at 0, has an integrable singularity at 1 and is $O(\log v)$ at infinity. The substitution $u = jv$ gives $\int_0^\infty \big|\log|1 - u^2/j^2|\big| u^{-\sigma-1} du = j^{-\sigma} K_\sigma$. Let $1 < \sigma < 2$. Then $\sum_j j^{-\sigma} K_\sigma < \infty$, the sum over j may be exchanged with the integral in (7.6), and

$$G(s) - A(s) = \zeta(s)\, J(s), \qquad J(s) = \int_0^\infty \log|1 - v^2|\, v^{-s-1}\, dv. \tag{7.8}$$

We compute $J(s)$ for $0 < \sigma < 2$. On $(0, 1)$, $\log(1 - v^2) = -\sum_{k\ge1} v^{2k}/k$ with terms of one sign, so $\int_0^1 \log(1 - v^2)\, v^{-s-1} dv = -\sum_k \frac{1}{k(2k-s)}$. On $(1, \infty)$ substitute $v = 1/w$: $\int_1^\infty \log(v^2 - 1)\, v^{-s-1} dv = \int_0^1 \big(\log(1 - w^2) - 2\log w\big) w^{s-1} dw = -\sum_k \frac{1}{k(2k+s)} + \frac{2}{s^2}$. Adding the two parts,

$$J(s) = \frac{2}{s^2} - \sum_{k\ge1} \frac{1}{k}\Big(\frac{1}{2k - s} + \frac{1}{2k + s}\Big) = \frac{2}{s^2} - \sum_{k\ge1} \frac{1}{k^2 - s^2/4}. \tag{7.9}$$

The partial fractions of the cotangent [5, (4.22.3)], at the point $\frac{\pi s}{2}$, which is not a multiple of π, read $\cot\frac{\pi s}{2} = \frac{2}{\pi s} - \frac{s}{\pi}\sum_{k\ge1} \frac{1}{k^2 - s^2/4}$. Multiplying by $\frac{\pi}{s}$ and comparing with (7.9) gives $J(s) = \frac{\pi}{s}\cot\frac{\pi s}{2}$. This proves (7.4) for $1 < \sigma < 2$. Both sides are analytic in the strip $0 < \sigma < 2$: G by Lemma 3.2, A by (7.2), and the right side because the simple pole of ζ at $s = 1$ is cancelled by the zero of the cotangent. So (7.4) holds in the strip. Near $s = 1$, $\cot\frac{\pi s}{2} = -\frac{\pi}{2}(s - 1) + O((s - 1)^3)$ and $\zeta(s) = \frac{1}{s-1} + O(1)$, which gives the value $-\frac{\pi^2}{2}$.

(c) For non-real s the number $\frac{\pi s}{2}$ is not real, so $\cot\frac{\pi s}{2}$ is finite and different from 0, and $\frac{\pi}{s} \ne 0$. By (7.4), $G(s) - A(s)$ is the product of $\zeta(s)$ and a finite non-zero number, so each of $G(s) = A(s)$ and $\zeta(s) = 0$ implies the other. □

By (7.6) the difference $G - A$ is the transform over all delays of h relative to its short-time form, and by (7.4) it is the Riemann zeta function up to an elementary factor. The function G carries the delays after the first return and A the delays before it. By Proposition 7.2(c) the two coincide at a non-real exponent of the strip exactly at the zeros of ζ. This equivalence is a consequence

of the explicit factor in (7.4) and gives no information on the location of those zeros. Mellin representations of ζ over a half-line are classical; the simplest, $\zeta(s) = \frac{s}{s-1} - s\int_1^\infty (u - \lfloor u \rfloor) u^{-s-1} du$ for $\sigma > 0$, follows for $\sigma > 1$ from $\zeta(s) = s\int_1^\infty \lfloor u \rfloor u^{-s-1} du$ and extends to $\sigma > 0$ because the integral is analytic there. The identity (7.4) is of the same type.

Figures 4 and 5 show $|A|$. On the grid of Figure 4 the computed values of $|A|$ lie between 0.0721 and 2.1831, and no zero of A is found there. Near the real axis the continuation (7.3) has the poles $0, 2, 4, \ldots$ and real zeros, of which Table 4 lists those in $-3.5 < s < 9.5$. Figure 6 shows, in the rectangle of Figure 3, the six computed zeros of G and the six points with $G = A$. The latter are the points $\frac{1}{2} + i\theta$ with $\theta = 14.1347$, 21.0220, 25.0109, 30.4249, 32.9351, 37.5862, the first six zeros of ζ in the upper half-plane.

TABLE 4. Computed zeros of G in the rectangle $0 < \operatorname{Re} s < 2$, $0 < \operatorname{Im} s \leq 40$, and computed real zeros of the continuation (7.3) of A in $-3.5 < s < 9.5$.

zeros of G	real zeros of A
$1.127313 + 15.001263\,i$	-2.0000
$0.533572 + 21.763255\,i$	-1.0000
$0.894731 + 25.624837\,i$	3.0428
$0.227688 + 30.975963\,i$	5.0981
$0.950507 + 33.599088\,i$	7.1478
$0.430900 + 38.054886\,i$	9.1892

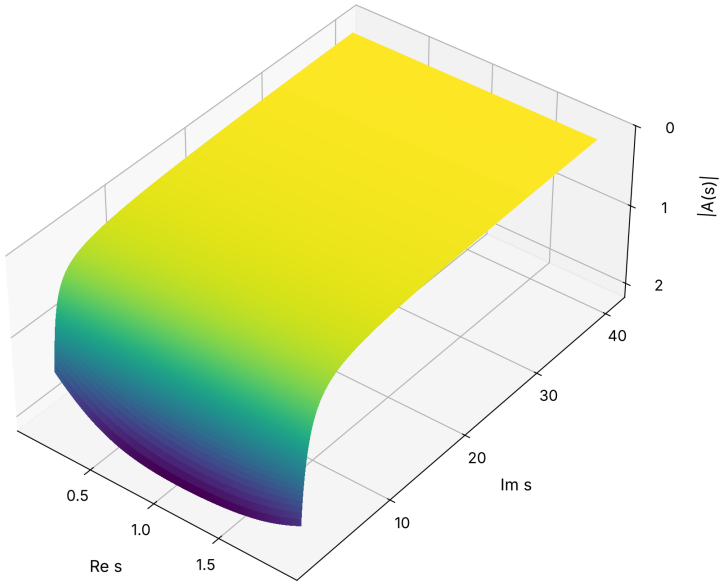


FIGURE 4. $|A(s)|$ on the rectangle of Figure 3, with the vertical axis inverted.

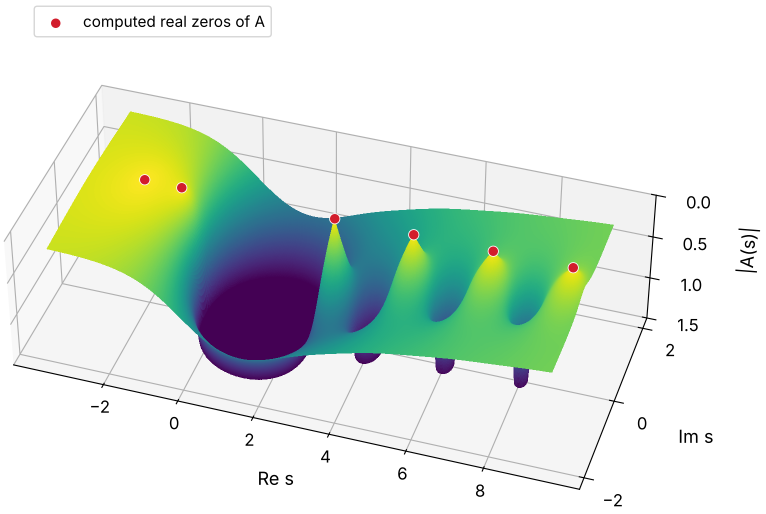


FIGURE 5. $|A(s)|$ near the real axis, computed from (7.3) and clipped above 1.5, with the vertical axis inverted. The red markers are the computed real zeros of A of Table 4; the wells are the poles $0, 2, 4, 6, 8$.

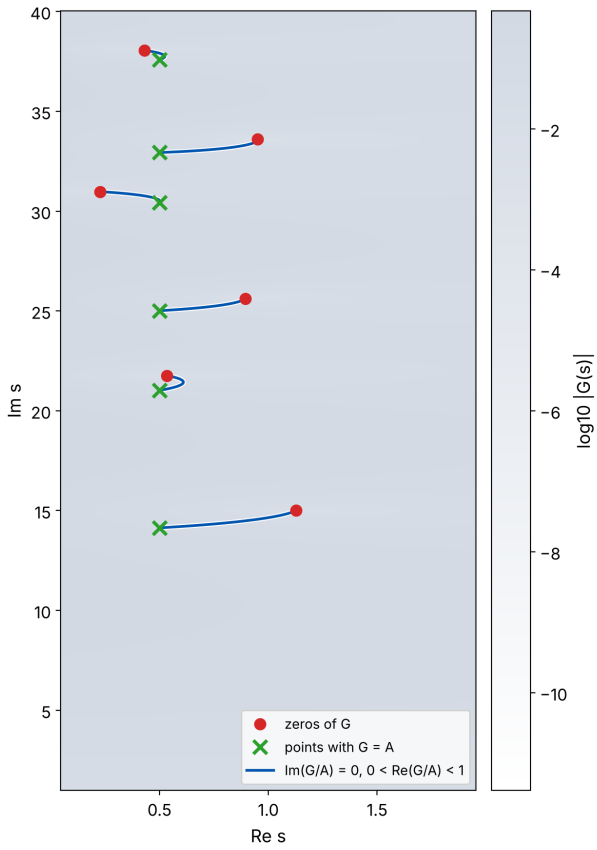


FIGURE 6. Computed zeros of G (dots), points with $G = A$ (crosses), and arcs of $\mathrm{Im}(G/A) = 0$ with $0 < \mathrm{Re}(G/A) < 1$. Each arc joins a zero of G to a point with $G = A$; this pairing is an observation on the computed data.

8. Conclusions

The main conclusion is the bound (4.4). For a two-level probe coupled dispersively to a thermal mode, the geometric mean of the Ramsey contrast after the first return, taken with the weight u^{-2}, satisfies $p_0 < \mathcal{V}_1 < \frac{1}{2}\exp(C_g)\,p_0$ for every positive occupation, where p_0 is the probability that the mode is empty and $C_g = 0.743794\ldots$ is the log-sine constant (3.14). The two constants 1 and $\frac{1}{2}\exp(C_g) = 1.05195\ldots$ cannot be improved. The log-sine constant is thus the sharp constant of a functional of the ideal contrast that can in principle be estimated from a sequence of Ramsey measurements: no thermal state gives a ratio $\mathcal{V}_1/p_0$ above $\frac{1}{2}\exp(C_g)$, and every smaller constant is exceeded at a sufficiently high occupation.

The second conclusion states what this functional measures that the half-period contrast does not. For every photon statistics of compound Poisson type, including thermal, Poisson and displaced thermal statistics, $p_0 < \mathcal{V}_1 \le p_0^{\,1-c_1(1)}$ with $c_1(1) = 0.0409\ldots$, so that the ratio $\log\mathcal{V}_1/\log p_0$ lies in $[1-c_1(1), 1)$ without a further hypothesis on the statistics; the half-period contrast is the modulus of the mean parity and, for a given p_0, can take any value between p_0^2 and 1 (Proposition 4.4). Within the thermal model the half-period contrast gives $\bar{n}$ exactly and more simply.

The third conclusion concerns damping. In the rate equations of the damped mode the coherence differs from the ideal one by at most $4\bar{n}(\bar{n}+1)\kappa t$ (Proposition 5.1). For the geometric contrast over $1 \le u \le U$ this requires κ/χ well below $1/\big(4\pi\bar{n}(\bar{n}+1)(2\bar{n}+1)U\big)$, a condition met at low occupation in long-lived modes and not at high occupation, and the half-period contrast is less affected (Corollary 5.2). The returns of the contrast measure the intrinsic decay of the probe and test the linearity of the shift.

Further conclusions follow from the same analysis. The bound holds for every power weight u^{-s-1} with $s > 0$, with the sharp constant $e^{-sG(s)}$, so that the function G on the positive real axis is the logarithm of the sharp constant divided by $-s$ (Theorem 4.1). For thermal states the occupation is bracketed by the geometric contrast alone, $1/\mathcal{V}_1 - 1 < \bar{n} < \exp(C_g)/(2\mathcal{V}_1) - 1$ (Corollary 4.2), and the geometric and half-period contrasts satisfy the relation (4.6), which contains no parameter and which coherent states of mean photon number at least 0.3 violate (Corollary 4.3). At high occupation $\mathcal{V}_1 = \frac{1}{2}\exp(C_g)p_0\,q^{\pi^2/12-3/4}e^R$ with $0 < R \le \frac{\tau^2}{\pi^2}(\frac{5}{2} + \frac{1}{2}\log\frac{1}{\tau})$ (Theorem 4.5). A finite range of delays $1 \le u \le U$ changes $\log\mathcal{V}_s$ by at most $U^{-s}\log(2\bar{n}+1)$ (Proposition 4.6). The integral of the logarithm of the contrast over all delays with the weight u^{-2} is exactly $-\pi^2\bar{n}$, and the first return separates it into a part within $C_g - \log 2$ of $\log p_0$ and a part that carries the linear term (Proposition 6.1). The product $\mathcal{P}(N)$ of the contrasts at the harmonic delays N/k satisfies $\mathcal{P}(N)^{1/N} \to \mathcal{V}_1$, with the explicit bound (6.4) for its logarithmic rate (Proposition 6.2).

At complex exponents G is the hot limit of the transform of $\log(V/p_0)$ after the first return, and its zeros are the weights on which that excess has no component (Proposition 7.1). Together with the transform A of the delays before the first return it satisfies $G - A = \frac{\pi}{s}\cot(\frac{\pi s}{2})\zeta$, so that in the strip $0 < \operatorname{Re} s < 2$ the two transforms coincide at a non-real exponent exactly at the zeros of the Riemann zeta function (Proposition 7.2). This is a consequence of an explicit factorisation and gives no information on the location of those zeros.

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